

## PROBLEM SET 9

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 10)

1) **(10 pts)** A particle is in a potential

$$V(x) = V_0 \sin(2\pi x/a)$$

which is invariant under the translations  $x \rightarrow x + ma$ , where  $m$  is an integer. Verify whether or not momentum is conserved.

If momentum is not, then why not ?